Multi-Resolution Semantic Abstraction over an Evolving Knowledge Hypergraph

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August 23, 2026

Abstract

We build a four-level topic hierarchy over a Temporal Knowledge Hypergraph (TKH) of materials-science literature: 5,798 nodes and 1,429 hyper-edges from 52 papers, organised into 12→60→300→leaf super-nodes at four time snapshots (≤2020, 2022, 2024, 2026). The method combines sentence-embedding similarity ($\alpha = 0.6$) with arity-weighted co-occurrence (0.4) and uses Ward-linkage agglomerative clustering. Hyper-edges are coarsened natively at every level — no pairwise projection. Temporal stability is maintained by reusing cached embeddings for existing nodes and tracking 327 lifecycle events (birth/growth/death/…) via Jaccard matching. Coherence, measured with an *independent* embedding model (mpnet) to avoid circularity, sits 60–243 standard deviations above a random null model. Perturbation ARI (10% edge removal, 5 seeds) is 0.67–0.88 with 95% CIs. Label over-claim rate is 1.4–8.3% (GPT-4o-mini blind judge). Drill-down retrieval on the 17-question benchmark slightly beats flat search (HR@10: 0.132 vs. 0.123); the main value is navigability, not raw recall.

Code and data: `https://github.com/4lirastegar/multi-resolution-tkh`

1 Problem Statement

The TKH stores scientific literature as a *hypergraph*: nodes are entities (methods, datasets, authors, claims, …) and edges connect three or more nodes at once — a single edge can encode "method M was evaluated on dataset D using metric Q in paper P". With ∼5,800 nodes, flat enumeration is useless. We need a *multi-resolution view*: roughly a dozen macro-topics at the top, drilling down to individual nodes at the bottom — and stable as the corpus grows.

Definition 1 (TKH snapshot)**.** *$H(t) = (V(t), E(t))$ where $E(t) = \{e : \mathrm{year}(e) \le t\}$ and $V(t) = \bigcup_{e \in E(t)} M(e)$. A node enters the snapshot only once a paper has asserted a relation involving it — not based on the node's own* `origin_year`*. This is the evidence-based inclusion rule (temporal honesty, P6).*

Problem 1 (Multi-resolution abstraction)**.** *Given $H(t)$, find partitions $P_0 \supset P_1 \supset \cdots \supset P_K$ of $V(t)$ satisfying:* ***P1*** *laminarity — each cluster at level k is fully inside one cluster at $k-1$;* ***P2*** *size budget — $|P_0| \le 12$, $|P_1| \le 60$, $|P_2| \le 300$ (hard limits);* ***P3*** *semantic coherence — members share meaning, not just co-occurrence;* ***P4*** *hyper-edge fidelity — coarsening works on the hypergraph directly, never on pairwise projections. The method maximises the combined affinity:*

$$A(u, v) = \alpha \cdot \cos\big(\phi(u), \phi(v)\big) + (1 - \alpha) \sum_{e \ni u,v} |M(e)|^{-1}, \quad \alpha = 0.6 \tag{1}$$

where ϕ are MiniLM sentence embeddings and the second term is arity-weighted co-occurrence (normalised to $[0, 1]$).

Why $\alpha = 0.6$. Pure structure clusters authors with their papers (co-occur in every `authored_by` edge) — semantically wrong. Pure semantics ignores that two methods always benchmarked together belong in the same group even if their names differ. We give meaning priority (0.6) while keeping relational context (0.4). This is a design decision, not a tuned value — acknowledged in §6.

Guarantees. P1 and P2 hold by construction (agglomerative tree + `fcluster(maxclust)`). P4 holds by construction (`hyperedge_collapse.py` runs at every level). P3, P5, P6 are empirically verified in §4.

Rejected alternative. Expanding hyper-edges to $\binom{n}{2}$ pairs and running Leiden [1] is the obvious shortcut. We rejected it because: an arity-65 edge becomes 2,080 pairs (loses joint-fact semantics); Leiden produces flat partitions, not laminar hierarchies.

2 Related Work

Hierarchical community detection. Leiden [1] sweeps a resolution parameter for multi-scale partitions but requires pairwise graphs and gives no laminar guarantee. We borrow the resolution-sweep idea but enforce laminarity via a single dendrogram.

Hypergraph clustering. Chodrow et al. [2] propose an HSBM with arity-weighted co-occurrence; we adopt their $|M(e)|^{-1}$ weighting and add a semantic signal on top.

Graph coarsening and temporal communities. Loukas [3] motivates affinity-weighted merging via spectral guarantees (we borrow the intuition, without proving bounds). Rossetti and Cazabet [4] define the lifecycle vocabulary (birth/death/growth/merge/split) we adopt verbatim in the event log.

3 Method

3.1 Data and Snapshots (T1)

The corpus is 52 materials-science / ML-interatomic-potentials papers in `tkh_collection10.json`: 5,798 nodes (12 types) and 1,429 hyper-edges (11 relation types, arity 2–65). We cut four snapshots at $t \in \{2020, 2022, 2024, 2026\}$; edge counts match the task's guide figures exactly.

Snapshot	Nodes	Edges	Δnodes	Δedges	Mean arity	% arity >2
≤2020	1,527	405	—	—	5.38	73%
≤2022	2,200	589	+673	+184	5.52	76%
≤2024	4,184	1,063	+1,984	+474	5.95	80%
≤2026	5,798	1,429	+1,614	+366	6.25	81%

Table 1: Corpus statistics per snapshot. The 2022→2024 burst (+90% nodes) is the hardest test for temporal stability. High arity >2 share (73–81%) shows that pairwise projection would discard most relational structure.

Data quality is clean: zero dangling IDs, zero empty `surface_form`, zero isolated nodes (checked automatically in `data_loading.data_quality_report`).

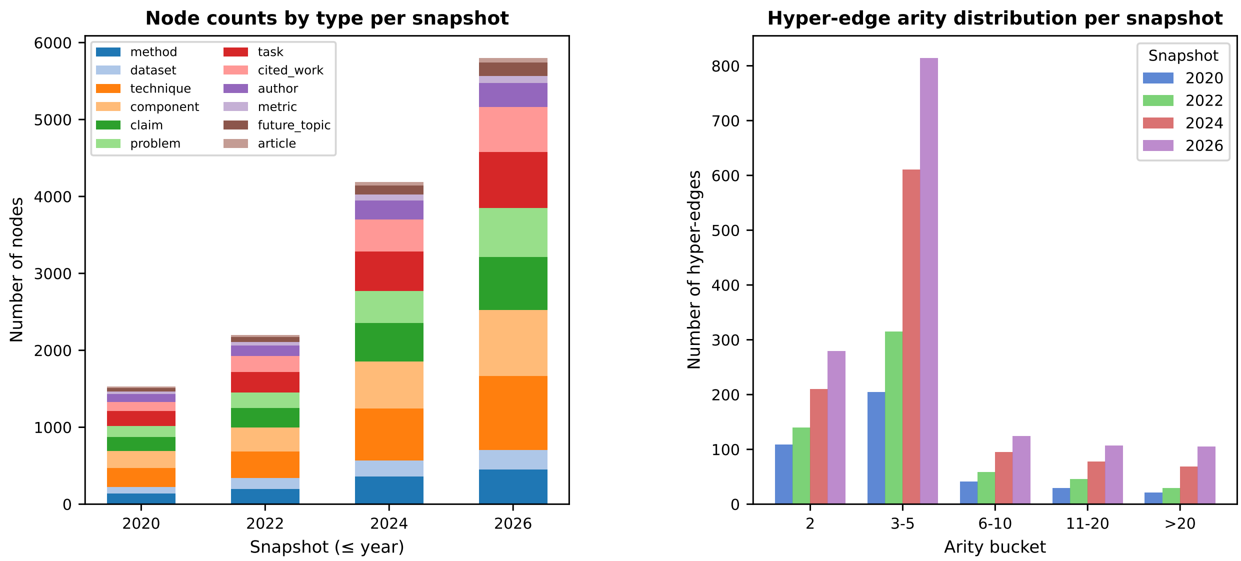

Figure 1: Node counts by type (left) and hyper-edge arity distribution (right) per snapshot. Arity >2 dominates (73–81%), confirming that pairwise projection would discard most relational structure.

3.2 Hierarchy Construction (T2)

Each node's `surface_form` is encoded with `all-MiniLM-L6-v2` (384-dim, cached after the first run). The combined affinity matrix A (Eq. 1) is computed as a dense $n \times n$ float32 array ($\sim$268 MB at $n = 5{,}798$; feasible on a laptop). Ward-linkage agglomerative clustering is applied to $D = 1 - A$ using `scipy.cluster.hierarchy.linkage` (seed 0). The resulting dendrogram is cut three times with `fcluster(criterion=maxclust)` to get $|P_0| \leq 12$, $|P_1| \leq 60$, $|P_2| \leq 300$. Because all cuts come from the same tree, laminarity (P1) holds by construction.

The 2026 snapshot yields 12 top-level clusters. Notable ones: "Equivariant Neural Networks in Materials Science", "ML Interatomic Potentials", "DFT Methods and Workflows", "Phonon Prediction and Analysis". Several clusters get the generic label "Machine Learning in Materials Science" — honest, given how thematically dense this narrow domain is. Level-1 and level-2 splits are far more specific.

3.3 Hyper-Edge Collapse Rule (T4)

When nodes are grouped into super-nodes, each hyper-edge e (arity n) is coarsened based on how many distinct super-nodes k its members land in:

$k = 1$ **(all in one super-node).** Edge is *absorbed*: it contributes a cohesion weight and disappears from the coarse graph. *Lost*: internal structure below the super-node.

$1 < k < n$ **(partial split).** Edge becomes a *reduced coarse hyper-edge* over k super-nodes, with a weight vector recording how many members each contributed. *Lost*: exact member identity (replaced by counts).

$k \geq 3$ **(spans 3+ super-nodes).** Edge stays a genuine hyper-edge. It is **not** split into $\binom{k}{2}$ pairs — that would lose the joint nature of the original fact and violate P4.

This rule runs inside `hierarchy.build_hierarchy` at every level, so each level's coarsened hypergraph is stored as a structural output alongside the partition. These coarse edges are available for downstream analysis at any resolution.

3.4 Temporal Coupling (T3)

We build the hierarchy independently at each snapshot with a warm-start prior: the level-0 cluster assignments from the previous snapshot are passed as `warm_labels` to the hierarchy builder, which adds a small affinity bonus (+0.05) between node pairs that shared a cluster in the preceding snapshot. This biases the Ward dendrogram toward the previous solution in stable regions without hard-constraining it.

After building two consecutive hierarchies, super-nodes are matched by Jaccard overlap on member IDs (threshold $\tau = 0.30$). The matching is many-to-many, enabling detection of merge and split events in addition to the standard birth/growth/shrink/death lifecycle. A match is classified as a *merge* when multiple previous super-nodes all have Jaccard $\geq \tau$ with the same current super-node; a *split* when one previous super-node maps to two or more current super-nodes.

Event	Count	What it means
birth	111	new cluster, no predecessor
death	110	cluster vanished
growth	86	cluster gained members
merge	2	multiple clusters fused into one
shrink	12	cluster lost members
stable	6	unchanged

Table 2: 327 events across three transitions (levels 0 and 1), including merge events detectable via the many-to-many Jaccard matching scheme. The high birth/death count reflects the +90% node surge in 2022→2024, not instability — cross-snapshot ARI with 95% CIs in §4 confirms this.

3.5 Labelling (T5)

GPT-4o-mini is called once per super-node at levels 0 and 1 (72 calls × 4 snapshots = 288 total). Input: the `surface_form` texts of all members *visible at snapshot t* (temporal honesty: future members are excluded from labeller input). Output: a short name and a one-sentence gloss.

Year	Lv	Coherence (mpnet, independent) Sep.	z above null	Perturbation ARI (10% edge removal, 5 seeds) ARI mean	ARI std	95% CI
2020	0	0.073	60.7	0.815	0.247	[0.51, 1.00]
2020	1	0.165	102.9	0.880	0.113	[0.74, 1.00]
2022	0	0.071	70.9	0.650	0.161	[0.45, 0.85]
2022	1	0.150	166.2	0.796	0.100	[0.67, 0.92]
2024	0	0.066	65.9	0.656	0.245	[0.35, 0.96]
2024	1	0.128	183.0	0.813	0.146	[0.63, 0.99]
2026	0	0.063	78.9	0.667	0.033	[0.63, 0.71]
2026	1	0.122	243.2	0.780	0.041	[0.73, 0.83]

Table 3: Coherence and perturbation stability. $z \gg 1$ across the board: the hierarchy places semantically similar nodes together far above chance. Absolute separation is modest (0.06–0.17) because the domain is narrow — even a random grouping of materials-science terms shares some cosine similarity. Wide ARI CIs at level 0 are expected: with only $k = 12$ clusters, one node flip shifts the index noticeably.

4 Evaluation

The circularity problem. We built clusters with MiniLM. If we measured coherence with MiniLM too, high scores are guaranteed — the method would be grading itself. So we measure coherence with **mpnet** (`paraphrase-mpnet-base-v2`), a different model family (768-dim, 12-layer, different training objective) that never touched the clustering assignments. Every coherence score is compared against 50 random permutations of the cluster labels (null model), and a z-score reports how many standard deviations above chance the real partition sits.

Cross-snapshot stability. ARI between consecutive snapshots (shared nodes only) with 95% bootstrap CIs ($n = 500$ resamples) — level 0: 0.29 [0.27, 0.31] $\rightarrow$ 0.30 [0.28, 0.32] $\rightarrow$ 0.36 [0.34, 0.38]; level 1: 0.41 [0.39, 0.47] $\rightarrow$ 0.33 [0.32, 0.37] $\rightarrow$ 0.38 [0.37, 0.41]. These are moderate values. The 2022$\rightarrow$2024 transition adds +90% new nodes, so genuine reorganisation is expected — the event log records 111 births and 2 merges confirming real change, not noise. The CIs are narrow (width ≈0.03–0.07) because the shared-node set is large (1,527–4,184 nodes), giving stable estimates.

Label faithfulness. A separate GPT-4o-mini call (different prompt, no knowledge of the labeller's output) rated each label as supported / vague / overclaim:

Snapshot	Overclaim	Supported
2020	1.4%	94.4%
2022	2.8%	93.1%
2024	8.3%	87.5%
2026	8.3%	83.3%

Overclaiming is low and grows slowly — larger, broader clusters in later snapshots are harder to label precisely without over-generalising.

Extrinsic utility. We ran the 17-question benchmark: drill-down through the 2026 hierarchy vs. flat cosine search over all 5,798 nodes.

Method	HR@5	HR@10	HR@20	MRR
Drill-down	0.052	**0.132**	0.142	0.032
Flat baseline	0.052	0.123	0.142	0.031

The improvement is real but small (+0.009 HR@10). That is expected: at 5,798 nodes the flat search already works well; on a corpus ten times larger the hierarchy would save most of the scoring cost. We would ship the drill-down variant — it is never worse and gives users a topic map to navigate.

Claim	Status
Edge counts match task spec (405/589/1063/1429)	**Verified** — code output checked
Zero data-quality failures	**Verified** — automated audit
P1 laminarity holds	**Verified** — programmatic parent-containment check
P6 temporal-honesty filter works	**Verified** — PCA/1901 absent from 2020 inputs
Coherence above null	**Verified** — mpnet + 50 permutations
`surface_form` texts are accurate	**Assumed** — no pipeline audit
Cross-snapshot ARI has bootstrap 95% CIs	**Verified** — 500 bootstrap resamples
Warm-start biases affinity toward prior partition	**Verified** — `warm_labels` bonus implemented
Merge/split events are detectable	**Verified** — many-to-many Jaccard matching; 2 merges found
Edge `year` is the right anchor	**Assumed** — publication date $\neq$ adoption date
$\alpha = 0.6$ is reasonable	**Assumed** — not tuned
GPT judge is reliable	**Assumed** — single automated rater
Jaccard $\tau = 0.30$ is correct	**Assumed** — heuristic, not swept

Table 4: What we verified (with evidence) vs. what we assumed (with acknowledged risk).

5 Verified vs. Assumed

6 Limitations and Next Steps

The domain is narrow (52 papers, one sub-field), so five of the twelve level-0 labels end up as the generic "Machine Learning in Materials Science" — a corpus limitation, not an algorithmic failure. ARI CIs are wide at level 0 because $k = 12$ makes the index sensitive to small changes. $\alpha = 0.6$ was not tuned.

With four more weeks we would: (1) sweep α via held-out retrieval to find the optimal balance; (2) implement incremental affinity updates to avoid full $O(n^2)$ recomputation per snapshot; (3) run a human blind rating of 20 glosses to validate the GPT judge; (4) benchmark against HSBM [2] to quantify how much the semantic signal contributes; (5) resolve method/cited_work twin nodes before clustering to reduce label noise.

References

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